

Chapter 1.4 Integration

Rick Durrett Probability: Theory and Examples

Hu Jingwu

Built from the local Chapter 1 LLMwiki flash cards.

Roadmap for 1.4

Learning goal

Understand the Lebesgue integral construction well enough to know why approximation arguments work.

1. Simple functions and canonical level-set sums.
2. Nonnegative integral as lower approximation.
3. Signed integrability through positive and negative parts.
4. Linearity, monotonicity, and a.e. replacement.
5. Approximation exercises: null sets, dyadic bins, L^1 density, and Riemann–Lebesgue.

Simple Functions

Definition

A measurable function ϕ is simple if it takes finitely many values. In canonical form,

$$\phi = \sum_{k=1}^n a_k \mathbf{1}_{A_k},$$

where the A_k are measurable level sets.

Integral

For $\phi \geq 0$,

$$\int \phi d\mu = \sum_{k=1}^n a_k \mu(A_k).$$

LLMwiki: Simple functions; ID: durrett_ch1_simple_functions

Simple-Function Proof Template

How to prove integral identities

Prove the statement first for indicators, then for nonnegative simple functions, then pass to limits.

Indicators are the atomic case:

$$\int \mathbf{1}_A d\mu = \mu(A).$$

Finite linear combinations follow by finite additivity on disjoint level sets. For general nonnegative functions, choose simple $\phi_n \uparrow f$ and use the definition of the nonnegative integral.

LLMwiki: Simple functions; ID: durrett_ch1_simple_functions

Integral of a Nonnegative Function

Definition

If $f \geq 0$ is measurable, define

$$\int f \, d\mu = \sup \left\{ \int \phi \, d\mu : 0 \leq \phi \leq f, \phi \text{ simple} \right\}.$$

Pitfall

The value may be $+\infty$. Do not subtract two infinite nonnegative integrals.

LLMwiki: Integral of a nonnegative function; ID: durrett_ch1_nonnegative_integral

Monotone Approximation Idea

Proof idea

The definition makes lower approximation the natural language for nonnegative integrals.

Given $f \geq 0$, construct simple functions ϕ_n with

$$0 \leq \phi_n \leq \phi_{n+1} \leq f, \quad \phi_n \uparrow f.$$

Then the integrals increase to $\int f d\mu$. This is the engine behind dyadic approximations and monotone convergence arguments.

LLMwiki: [Integral of a nonnegative function](#); ID: durrett_ch1_nonnegative_integral

Integrable Signed Functions

Definition

For a real measurable f , write

$$f^+ = \max(f, 0), \quad f^- = \max(-f, 0).$$

Then $f = f^+ - f^-$ and $|f| = f^+ + f^-$.

Integrability

We say f is integrable if

$$\int f^+ d\mu < \infty \quad \text{and} \quad \int f^- d\mu < \infty,$$

equivalently $\int |f| d\mu < \infty$.

Signed Integral: How to Define It

Definition

If f is integrable, set

$$\int f \, d\mu = \int f^+ \, d\mu - \int f^- \, d\mu.$$

The integrability assumption guarantees both terms are finite, so the subtraction is meaningful. This is exactly why one checks $\int |f| \, d\mu < \infty$ before using finite linearity or dominated convergence style arguments.

LLMwiki: Integrable signed functions; ID: durrett_ch1_integrable_function

Linearity and Order Properties

Theorem

For integrable functions, the integral is linear and monotone:

$$\int (af + bg) d\mu = a \int f d\mu + b \int g d\mu,$$

and if $f \leq g$ a.e., then

$$\int f d\mu \leq \int g d\mu.$$

If $f = g$ a.e., then $\int f d\mu = \int g d\mu$.

LLMwiki: Linearity and order properties of the integral; ID: durrett_ch1_integral_linearity

Linearity: How to Prove It

Start with indicators and disjoint simple functions, where the formula is finite algebra. Extend to nonnegative simple functions by canonical level sets. For general nonnegative functions, approximate from below by simple functions. Finally split signed functions into positive and negative parts, checking finiteness so no $\infty - \infty$ expression appears.

LLMwiki: [Linearity and order properties of the integral](#); ID: durrett_ch1_integral_linearity

Exercise 1.4.1

Show that if $f \geq 0$ and

$$\int f \, d\mu = 0,$$

then $f = 0$ almost everywhere.

LLMwiki: Zero integral of nonnegative function implies zero a.e.; ID:
`exercise_method_zero_integral_nonnegative`

Exercise 1.4.1: Proof Skeleton

For each $n \geq 1$, let

$$A_n = \{x : f(x) > 1/n\}.$$

Since $f \geq (1/n)\mathbf{1}_{A_n}$,

$$0 = \int f \, d\mu \geq \frac{1}{n}\mu(A_n),$$

so $\mu(A_n) = 0$. Finally,

$$\{f > 0\} = \bigcup_{n=1}^{\infty} A_n,$$

a countable union of null sets.

LLMwiki: Zero integral of nonnegative function implies zero a.e.; ID:
`exercise_method_zero_integral_nonnegative`

Exercise 1.4.2

Let $f \geq 0$ and

$$E_{n,m} = \left\{ x : \frac{m}{2^n} \leq f(x) < \frac{m+1}{2^n} \right\}.$$

Show that, as $n \rightarrow \infty$,

$$\sum_{m=1}^{\infty} \frac{m}{2^n} \mu(E_{n,m}) \uparrow \int f \, d\mu.$$

LLMwiki: [Dyadic lower simple approximations](#); ID: `exercise_method_dyadic_lower_integral_approximation`

Exercise 1.4.2: Proof Skeleton

Define

$$\phi_n(x) = \sum_{m=1}^{\infty} \frac{m}{2^n} \mathbf{1}_{E_{n,m}}(x).$$

Equivalently, for finite $f(x)$,

$$\phi_n(x) = 2^{-n} \lfloor 2^n f(x) \rfloor.$$

Hence $0 \leq \phi_n \leq f$ and $\phi_n \uparrow f$ pointwise. Also

$$\int \phi_n d\mu = \sum_{m=1}^{\infty} \frac{m}{2^n} \mu(E_{n,m}).$$

By monotone lower approximation for the nonnegative integral,

$$\int \phi_n d\mu \uparrow \int f d\mu.$$

Exercise 1.4.3

Let g be an integrable function on $\mathbb{R}$ and let $\varepsilon > 0$. Show, using the definition of the integral and interval approximation of measurable sets, that g can be approximated in L^1 by a continuous function. More explicitly:

1. approximate g by a simple function ϕ with $\int |g - \phi| dx < \varepsilon$;
2. approximate the sets in ϕ by finite unions of intervals to obtain a step function q with $\int |\phi - q| dx < \varepsilon$;
3. round the corners of q to obtain a continuous function r with $\int |q - r| dx < \varepsilon$.

LLMwiki: [L1 approximation by continuous functions](#); ID: `exercise_method_l1_continuous_approximation`

Exercise 1.4.3: Proof Skeleton

First approximate g by a simple function

$$\phi = \sum_{k=1}^N b_k \mathbf{1}_{A_k}, \quad \int_{\mathbb{R}} |g - \phi| dx < \varepsilon.$$

Approximate each measurable set A_k by a finite union of intervals U_k so well that

$$\sum_{k=1}^N |b_k| \lambda(A_k \triangle U_k) < \varepsilon.$$

Put $\psi = \sum_{k=1}^N b_k \mathbf{1}_{U_k}$ and refine endpoints to rewrite it as a step function q ; then

$$\int_{\mathbb{R}} |\phi - q| dx < \varepsilon.$$

Finally choose small boundary widths δ_j and replace each jump of q by a short linear ramp. The resulting r is continuous and can be chosen so that

$$\int_{\mathbb{R}} |q - r| dx < \varepsilon.$$

Thus $\int_{\mathbb{R}} |g - r| dx < 3\varepsilon$, and since ε is arbitrary this gives L^1 approximation.

LLMwiki: [L1 approximation by continuous functions](#); ID: `exercise_method_l1_continuous_approximation`

Exercise 1.4.4

Prove the Riemann–Lebesgue lemma: if g is integrable, then

$$\lim_{n \rightarrow \infty} \int_{\mathbb{R}} g(x) \cos(nx) \, dx = 0.$$

Hint: first prove it for step functions, then use the previous exercise.

LLMwiki: Riemann–Lebesgue by step-function reduction; ID:

`exercise_method_riemann_lebesgue_step_reduction`

Exercise 1.4.4: Proof Skeleton

First suppose

$$q(x) = \sum_{j=1}^M c_j \mathbf{1}_{(a_{j-1}, a_j)}(x).$$

Then

$$\int_{\mathbb{R}} q(x) \cos(nx) dx = \sum_{j=1}^M c_j \frac{\sin(na_j) - \sin(na_{j-1})}{n} \rightarrow 0.$$

Now choose a step function q with $\int_{\mathbb{R}} |g - q| dx < \varepsilon$. Since $|\cos(nx)| \leq 1$,

$$\left| \int g \cos(nx) dx \right| \leq \left| \int q \cos(nx) dx \right| + \varepsilon.$$

Taking lim sup and then letting $\varepsilon \downarrow 0$ proves the limit is 0.

[LLMwiki: Riemann-Lebesgue by step-function reduction](#); ID: `exercise_method_riemann_lebesgue_step_reduction`